XRPL Validator — Daily Engineering Report · 2026-09-21

Period: 2026-09-21 00:00 to 24:00 US Eastern (2026-09-21 04:00 to 2026-09-22 04:00 UTC); ledgers #107,128,672 to #107,151,004. Every count from the ledger and the validator's logs stops at #107,151,004. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,178 of the 22,333 ledgers the network closed (99.306%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 155 during planned restarts.
- **Native engine:** 0 engine receipts recorded.
- **Changes:** 2 deployments (cycles 147, 148), 14 commits merged into main, 1 engine finding fixed (F356).

Service levels

Measure	Value
Ledgers closed by the network	22,333
Validations issued	22,178
Coverage (validations / ledgers closed)	99.306%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	155
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	2
Basis: the hash the first peer announced as accepted (before cycle 183)	22,178

Restarts:

- 08:12:09 UTC: deployment of cycle 147; validations resumed at #107,132,666 (67 ledgers unsigned between the last validation before the restart and the first after it)
- 17:39:37 UTC: deployment of cycle 148; validations resumed at #107,141,474 (88 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,128,672–#107,151,004), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipts this day.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
147	`0aab032`	08:12:09 UTC	n/a	n/a	406 matched, 0 mismatched	#22 from 08:39 UTC: ended early: superseded by the next deployment (2026-09-21 18:06 UTC)
148	`c3904b9`	17:39:37 UTC	n/a	n/a	404 matched, 0 mismatched	#23 from 18:06 UTC: ended early: superseded by the next deployment (2026-09-22 13:52 UTC)

Cycle 147 delivered: `63e2a78` vectors: pin TicketCreate (28 tickets) and OfferCancel of a live offer — the two common mainnet types the gate had no byte-exact pin for; `840c268` vectors: pin UNLModify disable + re-enable (with the flag-ledger NegativeUNL rotation applied first) — the pseudo-transactions had no byte-exact pin; `9b3a153` vectors: pin EnableAmendment (fixCleanup3_3_0 activation, #106911489); pseudo drills move to pseudo_tx_vector.rs; `20bc9ec` vectors: pin EnableAmendment tfGotMajority (#107003649) — the Majorities path of the pseudo-transaction; `df80bb7` Finding 345: CredentialAccept judges the subject's reserve for the object it takes over (tecINSUFFICIENT_RESERVE); `b3ccc4d` Finding 346: a second CredentialAccept is tecDUPLICATE (lsfAccepted); harness: hydrate the DID object for DIDSet/DIDDelete; `64645e6` vectors: port-only suite — four campaign-6 payments (limit quality, partial+DeliverMin over paths) the port gets byte-exact and the model does not; `df4b355` Finding 347: AMMDelete is tecAMM_NOT_EMPTY while LP tokens are outstanding, terNO_AMM without a pool; the apply is deleteAMMAccount; `3a54c24` vectors: two more port-only campaign-6 payments (partial over a fee issuer; default-path EUR→USD); `fc54c49` vectors: seventh port-only campaign-6 payment (partial+DeliverMin, default paths, fee issuer); `0aab032` harness: the probe hydrates the AMM pre-state for AMMDelete too (finding 347's tec reads the pool object)

Cycle 148 delivered: `c6b4fd3` fixCleanup3_4_0: port the amendment's transactor-side rules behind the ledger's Amendments gate; `7fbf877` Findings 348-349: CheckCash judges the casher's line (auth, freeze) after the funds test; credentials::valid runs in EscrowFinish, PaymentChannelClaim, AccountDelete; `d74d5b3` Finding 350: a pseudo-account is created with Sequence 0 once SingleAssetVault or LendingProtocol is enabled (devnet #5488138); mainnet keeps the ledger sequence until then; `9428cef` amm: Finding 350 marker capitalised for the deploy marker grep; `a1cdddf` Batch: seven campaign-8 devnet vectors (all four modes, failures at every position, all-revert, multi-account BatchSigners, offer create+cancel); an AllOrNothing revert files no inner. Finding 351: Clawback's preclaim order; `79158d3` Finding 352: AMMClawback ported from rippled's applyGuts and equalWithdrawMatchingOneAmount; harness hydrates the holder's asset lines; `5c204d8` harness: the probe hydrates the holder's line for Clawback (its whole preclaim read-set); `faf89c6` Finding 353: EscrowFinish verifies the PREIMAGE-SHA-256 crypto-condition — tecCRYPTOCONDITION_ERROR for a wrong fulfillment, a missing or different Condition, or a Condition on an unconditioned escrow; `a05696d` harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots; `f48d696` harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots; `d4b5158` Finding 354: AccountDelete removes DID, Oracle, Delegate and Credential objects (rippled's nonObligationDeleter); harness hydrates the whole owner directory for AccountDelete; `0787019` Finding 355: AccountSet cannot clear GlobalFreeze under NoFreeze, and NoFreeze has no clear at all — both silent no-ops (SetAccount.cpp:346-375); nine campaign-12 TrustSet/AccountSet vectors; `21de443` test(node): campaign 13 — 15 testnet NFT depth vectors + NFT read-set hydration; `c3904b9` fix(ledger): F356 — EscrowFinish honours the destination's DepositAuth; campaign 14 vectors

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
08:56 UTC	xrpl-validator	`c6b4fd3`	fixCleanup3_4_0: port the amendment's transactor-side rules behind the ledger's Amendments gate
11:56 UTC	xrpl-validator	`7fbf877`	Findings 348-349: CheckCash judges the casher's line (auth, freeze) after the funds test; credentials::valid runs in EscrowFinish, PaymentChannelClaim, AccountDelete
12:08 UTC	xrpl-validator	`d74d5b3`	Finding 350: a pseudo-account is created with Sequence 0 once SingleAssetVault or LendingProtocol is enabled (devnet #5488138); mainnet keeps the ledger sequence until then
12:09 UTC	xrpl-validator	`9428cef`	amm: Finding 350 marker capitalised for the deploy marker grep
13:05 UTC	xrpl-validator	`a1cdddf`	Batch: seven campaign-8 devnet vectors (all four modes, failures at every position, all-revert, multi-account BatchSigners, offer create+cancel); an AllOrNothing revert files no inner. Finding 351: Clawback's preclaim order
13:50 UTC	xrpl-validator	`79158d3`	Finding 352: AMMClawback ported from rippled's applyGuts and equalWithdrawMatchingOneAmount; harness hydrates the holder's asset lines
13:54 UTC	xrpl-validator	`5c204d8`	harness: the probe hydrates the holder's line for Clawback (its whole preclaim read-set)
14:16 UTC	xrpl-validator	`faf89c6`	Finding 353: EscrowFinish verifies the PREIMAGE-SHA-256 crypto-condition — tecCRYPTOCONDITION_ERROR for a wrong fulfillment, a missing or different Condition, or a Condition on an unconditioned escrow
14:19 UTC	xrpl-validator	`a05696d`	harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots
14:20 UTC	xrpl-validator	`f48d696`	harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots
14:54 UTC	xrpl-validator	`d4b5158`	Finding 354: AccountDelete removes DID, Oracle, Delegate and Credential objects (rippled's nonObligationDeleter); harness hydrates the whole owner directory for AccountDelete
15:27 UTC	xrpl-validator	`0787019`	Finding 355: AccountSet cannot clear GlobalFreeze under NoFreeze, and NoFreeze has no clear at all — both silent no-ops (SetAccount.cpp:346-375); nine campaign-12 TrustSet/AccountSet vectors
15:50 UTC	xrpl-validator	`21de443`	test(node): campaign 13 — 15 testnet NFT depth vectors + NFT read-set hydration
16:43 UTC	xrpl-validator	`c3904b9`	fix(ledger): F356 — EscrowFinish honours the destination's DepositAuth; campaign 14 vectors

Findings fixed

- **fix(ledger): F356 — EscrowFinish honours the destination's DepositAuth; campaign 14 vectors** (`c3904b9`, xrpl-validator) Finding 356 (campaign 14, testnet CFD60CB8BA7A): rippled's EscrowFinish calls verifyDepositPreauth right after the tecNO_DST test (EscrowFinish.cpp:316) — a destination under lsfDepositAuth accepts the finish only from itself, from an account it preauthorized, or through credentials naming a DepositPreauth object; otherwise tecNO_PERMISSION. Our EscrowFinish never asked and paid the escrow out.

Other engineering changes

- **fixCleanup3_4_0: port the amendment's transactor-side rules behind the ledger's Amendments gate** (`c6b4fd3`, xrpl-validator) Ported (each gated on amendments::fix_cleanup_3_4_0, unit-tested; no oracle exists — the vendored libxrpl predates the amendment and testnet has it off): - EscrowFinish/EscrowCancel: the escrow's reserve is recycled before the delivery/return re-creates a line (EscrowFinish.cpp:345, EscrowCancel.cpp:174); EscrowCancel also gains the pre-existing createAsset reserve check. - OfferCreate + Payment DomainID: …

Tests and campaigns

- **test(node): campaign 13 — 15 testnet NFT depth vectors + NFT read-set hydration** (`21de443`, xrpl-validator)

Docs and operations

- **Findings 348-349: CheckCash judges the casher's line (auth, freeze) after the funds test; credentials::valid runs in EscrowFinish, PaymentChannelClaim, AccountDelete** (`7fbf877`, xrpl-validator)
- **Finding 350: a pseudo-account is created with Sequence 0 once SingleAssetVault or LendingProtocol is enabled (devnet #5488138); mainnet keeps the ledger sequence until then** (`d74d5b3`, xrpl-validator)
- **amm: Finding 350 marker capitalised for the deploy marker grep** (`9428cef`, xrpl-validator)
- **Batch: seven campaign-8 devnet vectors (all four modes, failures at every position, all-revert, multi-account BatchSigners, offer create+cancel); an AllOrNothing revert files no inner. Finding 351: Clawback's preclaim order** (`a1cdddf`, xrpl-validator)
- **Finding 352: AMMClawback ported from rippled's applyGuts and equalWithdrawMatchingOneAmount; harness hydrates the holder's asset lines** (`79158d3`, xrpl-validator)
- **harness: the probe hydrates the holder's line for Clawback (its whole preclaim read-set)** (`5c204d8`, xrpl-validator)
- **Finding 353: EscrowFinish verifies the PREIMAGE-SHA-256 crypto-condition — tecCRYPTOCONDITION_ERROR for a wrong fulfillment, a missing or different Condition, or a Condition on an unconditioned escrow** (`faf89c6`, xrpl-validator)
- **harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots** (`a05696d`, xrpl-validator)
- **harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots** (`f48d696`, xrpl-validator)
- **Finding 354: AccountDelete removes DID, Oracle, Delegate and Credential objects (rippled's nonObligationDeleter); harness hydrates the whole owner directory for AccountDelete** (`d4b5158`, xrpl-validator)
- **Finding 355: AccountSet cannot clear GlobalFreeze under NoFreeze, and NoFreeze has no clear at all — both silent no-ops (SetAccount.cpp:346-375); nine campaign-12 TrustSet/AccountSet vectors** (`0787019`, xrpl-validator)

Independence program

No independence record for UTC day 2026-09-21; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- PermissionDelegationV1_1 gained majority — ledger 107,135,489, 2026-09-21 11:18 UTC

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC
PermissionDelegationV1_1	2026-09-21 11:18 UTC	2026-10-05 11:18 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **AMM Self-Referential Volatility Fee Bounds** (amm-defi) Add sfFeeMin/sfFeeMax bounds and a bounded 8-entry sfPriceHistory circular buffer (STArray of {sfLedgerSeq, sfPrice}) to the AMM ledger object, settab
- **On-Ledger Finality Certificate Pseudo-Transaction** (consensus) Add a new pseudo-transaction ttFINALITY_CERT, injected by validators at ledger close in the same class as the existing NegativeUNL ttUNL_MODIFY, that
- **Spend-Policy Enforcement on PayChan Claims** (custody) Extend the PayChanClaim transactor's preclaim/doApply to consult the claimant's AccountPolicy object (g18's owner-reserved ltACCOUNT_POLICY, keylet::a

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-21 08:56	09-21 08:56	xrpl-validator	`c6b4fd3`	fixCleanup3_4_0: port the amendment's transactor-side rules behind the ledger's Amendments gate
09-21 11:56	09-21 11:56	xrpl-validator	`7fbf877`	Findings 348-349: CheckCash judges the casher's line (auth, freeze) after the funds test; credentials::valid runs in EscrowFinish, PaymentChannelClaim, AccountDelete
09-21 12:08	09-21 12:08	xrpl-validator	`d74d5b3`	Finding 350: a pseudo-account is created with Sequence 0 once SingleAssetVault or LendingProtocol is enabled (devnet #5488138); mainnet keeps the ledger sequence until then
09-21 12:09	09-21 12:09	xrpl-validator	`9428cef`	amm: Finding 350 marker capitalised for the deploy marker grep
09-21 13:05	09-21 13:05	xrpl-validator	`a1cdddf`	Batch: seven campaign-8 devnet vectors (all four modes, failures at every position, all-revert, multi-account BatchSigners, offer create+cancel); an AllOrNothing revert files no inner. Finding 351: Clawback's preclaim order
09-21 13:50	09-21 13:50	xrpl-validator	`79158d3`	Finding 352: AMMClawback ported from rippled's applyGuts and equalWithdrawMatchingOneAmount; harness hydrates the holder's asset lines
09-21 13:54	09-21 13:54	xrpl-validator	`5c204d8`	harness: the probe hydrates the holder's line for Clawback (its whole preclaim read-set)
09-21 14:16	09-21 14:16	xrpl-validator	`faf89c6`	Finding 353: EscrowFinish verifies the PREIMAGE-SHA-256 crypto-condition — tecCRYPTOCONDITION_ERROR for a wrong fulfillment, a missing or different Condition, or a Condition on an unconditioned escrow
09-21 14:19	09-21 14:19	xrpl-validator	`a05696d`	harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots
09-21 14:20	09-21 14:20	xrpl-validator	`f48d696`	harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots
09-21 14:53	09-21 14:54	xrpl-validator	`d4b5158`	Finding 354: AccountDelete removes DID, Oracle, Delegate and Credential objects (rippled's nonObligationDeleter); harness hydrates the whole owner directory for AccountDelete
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09-21 16:43	09-21 16:43	xrpl-validator	`c3904b9`	fix(ledger): F356 — EscrowFinish honours the destination's DepositAuth; campaign 14 vectors

Times in UTC. A commit appears on the day it was committed to main or merged into it.